# MINI INVERTER

A compact system that converts battery power into AC power

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Our mini inverter prototype

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INTRODUCTION

# What is a Mini Inverter?

A mini inverter is a compact power-conversion device. It takes DC electricity from a battery and converts it into AC electricity for suitable appliances.

### DC INPUT

The battery stores electrical energy as direct current (DC).

### CONVERSION

The inverter circuit switches and transforms the electrical energy.

### AC OUTPUT

The output socket can supply AC to a compatible load, if the design is rated for it.

**Main idea: Battery (DC) → Inverter circuit → AC output**

Prototype Overview

# Our Mini Inverter – Front View



## OUTPUT SOCKET

Connection point for a compatible AC load.

## ON / OFF SWITCH

Used to switch the unit on or off.

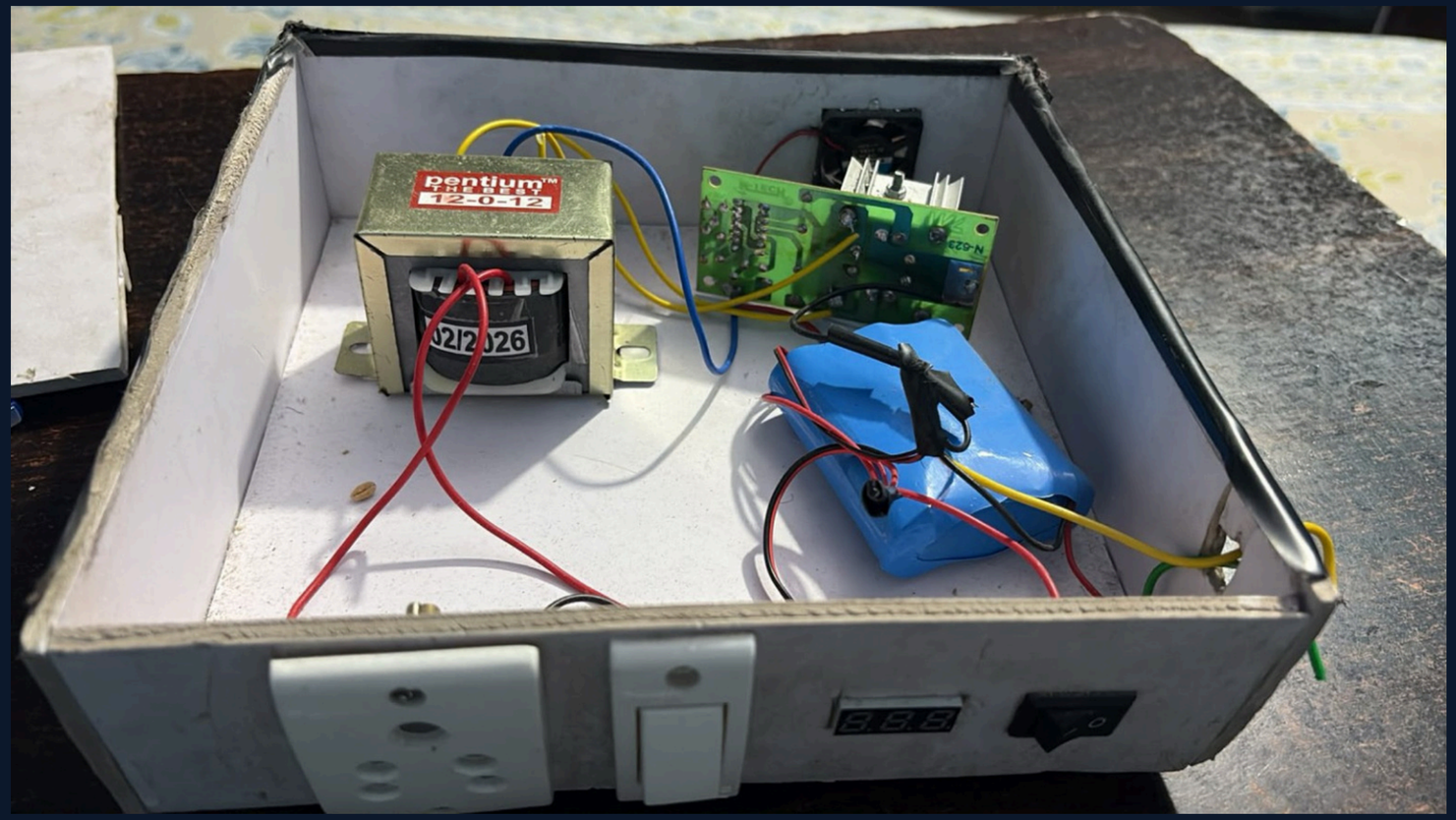
## DIGITAL DISPLAY

Likely a voltage/status display; exact reading depends on its wirin

The white switch beside the socket may be a separate control. Confirm its function from the circuit wiring.

Main visible part

# Inside the Prototype




### 12-0-12 TRANSFORMER

Visible transformer label suggests a 12-0-12 winding.

### BATTERY PACK

Stores energy and provides the DC supply.

### CONTROL PCB

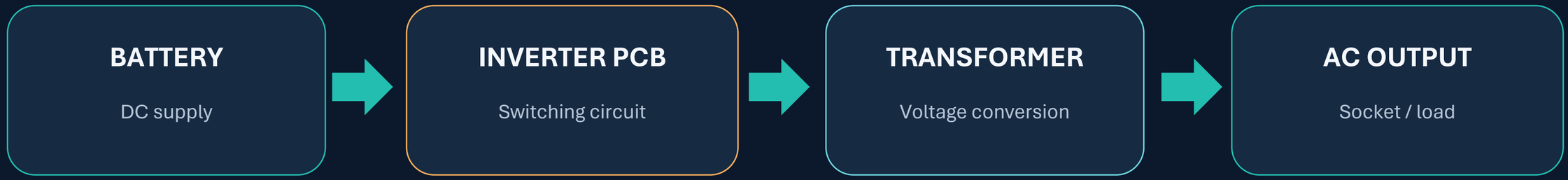
Contains the electronic switching/control components.

### COOLING FAN

Helps remove heat from the electronics.

Energy flow

# Basic Working Principle




The battery provides DC. The electronic circuit rapidly switches this power to create an alternating waveform. The transformer may then adjust the voltage before it reaches the output.

**Note: This is a simplified block diagram. The exact circuit sequence depends on the PCB design.**

Component details

# Function of Each Component

### Battery

Provides stored DC energy to run the inverter.

### Transformer

Transfers energy between windings and can step voltage up or down.

### PCB / Switching devices

Controls the switching process that produces AC from DC.

### Output socket

Allows a suitable appliance to be connected.

### Switch & display

Provide user control and a visual indication (function to be verified).

### Cooling fan

Moves air to help prevent overheating.

Where cant it be useful ?

# Applications & Advantages

## Possible applications

Small backup loads such as a suitable LED light, router, or other low-power device—only if the inverter rating supports it.

## Advantages

Compact size • Portable design • Demonstrates DC-to-AC conversion • Useful learning project

### A mini inverter is useful for understanding:

**POWER CONVERSION** — How electrical energy changes from DC to AC.

**BASIC ELECTRONICS** — How a PCB, transformer and battery work together.

**PRACTICAL DESIGN** — Why cooling, insulation and correct ratings matter.

Use Responsibly

# Safety & Important Limitations

### Mains-voltage danger

The output may be hazardous even though the battery is low voltage. Never touch exposed wires or terminals.

### Do not overload

Connect only loads within the verified power rating. Overloading can overheat or damage the unit.

### Battery precautions

Use the correct battery voltage and chemistry. Prevent short circuits and use suitable protection.

### Enclosure & ventilation

Keep wiring insulated, secure the enclosure, and do not block the cooling fan.

**Important: The photos do not show the full ratings or protection circuit. Do not assume the output is 230 V or connect household appliances until a qualified person verifies the design and tests it safely.**

Summary

# Conclusion

**This mini inverter prototype demonstrates the basic idea of converting battery DC power into AC power using electronic switching and a transformer.**

**INPUT**

Battery – DC energy

**PROCESS**

PCB + transformer

**OUTPUT**

AC at the socket

**Thank you! | Any questions?**